Image Module – Multimedia Security

Overview

The `Image.py` script provides core image processing and watermarking utilities for multimedia security applications. It supports several methods of watermark embedding and extraction (LSB, DCT, DWT, text-based and overlay techniques), as well as preprocessing tools for geometric attacks, binarization, and watermark visualization.

Key Functionalities

1. Text Conversion Utilities

- **`str_to_bitarray(text: str)`**
- **`bitarray_to_str(bit_array: np.ndarray)`**

- Encodes and decodes a UTF-8 string into a NumPy array of binary bits.

2. Image Preprocessing

- **`binarize(image_pil, threshold, mode)`**
  - Converts an image to a binary (black and white) version using a threshold.

- **`apply_geometric_attacks(image_pil, params)`**
  - Applies transformations such as scaling, rotation, resizing, and cropping to simulate geometric distortions.

3. Watermarking - Visual (Perceptual)

- **`apply_text_watermark(image_pil, params)`**
  - Adds a visible text watermark with user-defined location, size, color, and boldness.

- **`apply_overlay_watermark(image_pil, watermark_pil, method, key)`**
  - Applies an overlay watermark using additive, multiplicative, or transparency-based blending.

4. LSB-Based Watermarking

- **`embed_lsb_watermark(image_pil, watermark_pil, layer, bit_index)`**
  - Embeds a binary watermark image into the LSB of a selected color channel (RGB or HSV).

5. Frequency Domain Watermarking

- **`embed_dct_watermark(image_pil, watermark_pil, alpha, block_size)`**
  - Embeds watermark image into mid-band DCT coefficients block-by-block.

- **`embed_dwt_watermark(image_pil, watermark_pil, alpha, level)`**
  - Embeds watermark in the approximation coefficients of a Haar DWT transformation.

Importance in Project

The `Image.py` module enables:
- Robust watermarking against compression and attacks.
- Copyright assertion and verification through imperceptible or visible marks.
- Tamper detection by comparing altered vs original frequency domains.

Its modularity makes it easy to integrate in GUI systems or automation pipelines, and supports both covert (LSB, DCT, DWT) and overt (text, overlay) watermarking.

Dependencies

- cv2
- numpy
- pywt
- PIL

Video Module – Multimedia Security

Overview

The `Video.py` script is a comprehensive module designed for video processing and watermarking in multimedia security systems. It includes functions to validate and extract properties of video files, apply DCT and DWT watermarking methods, and export processed video. It also provides insights into audio embedded within the video.

Key Functionalities

1. Video Validation and Property Extraction

- **`validate(path: str)`**
  - Checks if a video file can be opened successfully.
  - Returns a capture object and a boolean status.

- **`get_properties(path: str, cap: cv2.VideoCapture)`**
  - Extracts video and audio properties such as:
    - Resolution
    - Frame count
    - FPS
    - Duration
    - Color mode
  - Audio information is extracted using `moviepy`, including:
    - Presence
    - Duration
    - Sample rate
    - Channels
    - Bit depth
    - Peak amplitude

2. Property Formatting

- **`format_properties(info: dict)`**
  - Returns a formatted string containing video and audio details.

3. Watermarking Techniques

- **`apply_dct_watermark(video_path, watermark_path, alpha: float)`**
  - Embeds a watermark image into each video frame using Discrete Cosine Transform (DCT).
  - The Y channel in YCrCb color space is modified with watermark data.

- **`apply_dwt_watermark(video_path, watermark_path, alpha: float)`**
  - Embeds a watermark image using Discrete Wavelet Transform (DWT).
  - The LL band of the grayscale frame is modified.

4. Watermarked Video Export

- **`export_video(path: str, video_capture: cv2.VideoCapture)`**
  - Saves frames from a `cv2.VideoCapture` object into a video file.
  - Maintains original resolution and frame rate.

5. Audio and Video Extraction.

Importance in Project

This module forms the backbone of video handling in a multimedia security suite. By integrating DCT and DWT watermarking algorithms, it ensures secure embedding of ownership or authentication data in video files. The inclusion of audio analysis provides an additional layer of metadata verification and forensic capability.

Dependencies

- `cv2`
- `numpy`
- `pywt`
- `moviepy`
- `tempfile`
- `os`

Audio Module – Multimedia Security

Overview

The `Audio.py` script is a utility module designed for audio processing, recording, playback, visualization, and watermarking. It supports operations such as reading audio files, recording live audio, playing audio, plotting waveforms, analyzing audio properties, and performing Least Significant Bit (LSB) watermark embedding and extraction. This module is a critical part of a multimedia security system.

Key Functionalities

1. Audio Input/Output

- **`read_audio(path: str)`**
  - Reads a `.wav` audio file.
  - Converts to 16-bit PCM if necessary.
  - Supports both mono and stereo audio.
  - Returns a tuple containing audio data, sample rate, and number of channels.

- **`export_audio(path: str, audio_data: np.ndarray, sample_rate: int, channels: int)`**
  - Saves audio data to a `.wav` file using 16-bit PCM encoding.

2. Audio Recording & Playback

- **`record_audio(duration: float, sample_rate: int, channels: int)`**
  - Records audio for the specified duration using the system microphone and return it as numpy ndarray.

- **`play_audio(audio_data: np.ndarray, duration: int, sample_rate: int)`**
  - Plays back the specified portion of the recorded or imported audio.

3. Audio Visualization

- **`plot_audio(audio_data: np.ndarray, sample_rate: int)`**

4. Audio Analysis

- **`get_properties(audio_data: np.ndarray, sample_rate: int, channels: int)`**
  - Extracts key audio characteristics such as:
    - Duration.
    - Sample Rate.
    - Channels.
    - Bit Depth.
    - Total Samples.

- Peak Amplitude.
- RMS Amplitude.
- Mean, Min, and Max Amplitudes.

5. LSB-Based Watermark Embedding

- **`embed(audio: np.ndarray, watermark: str, bit_idx: int)`**
  - Embeds a text watermark into audio data using Least Significant Bit substitution.

6. LSB-Based Watermark Extraction

- **`extract(audio: np.ndarray, bit_idx: int, length: int)`**
  - Extracts a text watermark from audio data based on the specified bit index and length.

Importance in Project

This module acts as the audio engine for a multimedia security application. It facilitates not only basic audio handling (read, record, play, plot) but also introduces basic steganographic techniques through LSB watermarking. The watermarking feature is crucial for:

Dependencies

- `wave`
- `numpy`
- `scipy.io.wavfile`
- `matplotlib`